



Have open borders contributed to European productivity
Growth? – An Economic impact Assessment of the
Schengen agreement 30 years later

ECO433 Empirical Strategies for Causal Analysis

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ABSTRACT

For the last 30 years, the Schengen agreement has ensured free movement across intra-European borders, simplifying cross-border commuting and the transportation of goods between countries. We estimate that the agreement has led to a 2.4% (~0.024 log point) increase in GDP per worker in Western Europe in the period 1995 – 2007. We furthermore discuss the implications of the recent re-introductions of border controls as a response to refugee flows and argue that they are likely to affect both the productivity and competitiveness of the already struggling continent negatively.

INTRODUCTION

2025 marks the 30th anniversary of the implementation Schengen agreement, and the 40th anniversary of its signing back in 1985. The essence of the agreement is that it allows citizens of participating nations to cross borders of the other member states without formal border controls, thus greatly simplifying international travel for both tourists, but crucially, for goods and services too. This has greatly altered living and traveling patterns of many Europeans, with numbers from the European Commission showing that 3.5 million people cross a Schengen border daily (European Commission, 2025a).

By increasing the mobility of both labor and goods, the agreement enables the establishment of intricate multinational value chains across Europe, thereby increasing competitiveness of European firms and businesses. For example, individuals can easily commute cross-border, thus making it easier for firms to mobilize labor with the required skills and talents regardless of which country the firm operates in (European Commission, 2025a).

The effects of the agreement on European trade have been widely studied, with positive effects on outcomes such as increase in demand for foreign-made goods and lowering the risk associated with trade in a foreign country (Davis & Gift, 2014). The authors furthermore found that European nations became closer trading partners as a result of the agreement.

There is however very little evidence linking the implementation of the Schengen agreement to increased productivity levels across Europe. In this paper, we utilize a Difference-in-Difference (DiD) approach to make a causal assessment of potential effects on growth in labor productivity. Our findings will also be relevant in the context of the discourse around the decline in European competitiveness and productivity as highlighted by the 2024 Draghi Report (Draghi, 2024).

THEORETICAL BACKGROUND AND RELATED LITERATURE

In this section, we provide a brief discussion of the theoretical effects of trade liberalization from the field of international trade and welfare theory. Nearly all literature from the field predicts that lowering trade barriers leads to increases in welfare and total utility for a given economy. The intuition is that

lowering the cost of trade between countries enables the exploitation of comparative advantages, with all nations participating in trade having the option of unbundling consumption from production, allowing each country to specialize in the field where it maintains a comparative advantage. This is illustrated below:

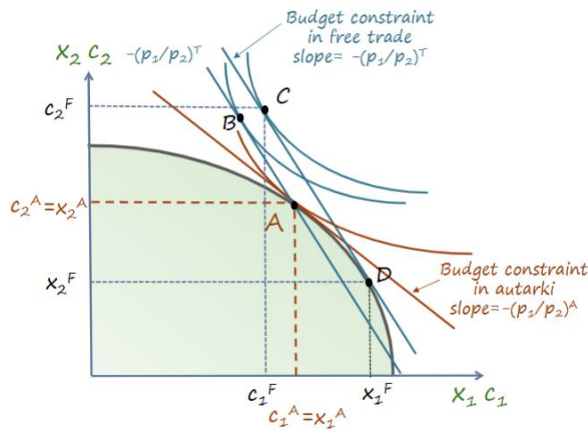


Figure 1: Utility gains from unbundling production and consumption (Orvedal, 2025)

The figure shows that when trade is possible, this given economy can specialize in the production of good 1 (denoted on the X-axis) and instead import its desired amount of good 2 (denoted on the Y-axis). Production of good 1 therefore increases, and the production of goods 2 falls, resulting in a positive shift in the utility curve.

Furthermore, Heterogenous-firm models predict that lowering friction between markets makes low-productivity firms shrink or exit the market, while high-productivity firms expand at their cost (Melitz and Redding, 2014). In theory, this should increase aggregate productivity even if there are no changes to in-firm innovation. The Schengen agreement in this case functions as a reduction in non-tariff friction in the form of easier transportation of goods and increased labor mobility. We therefore predict a post-entry increase in GDP per worker in our analysis.

Although there is little existing literature linking openness to increases in productivity, research shows that trade liberalization may have a positive effect on productivity growth through intertemporal knowledge spillovers in R&D (Gustafsson & Segerström, 2010). We therefore hypothesize that the Schengen agreement has had a positive effect on European productivity. In the following section, we provide a framework which we will subsequently utilize to assess the effects of the Schengen agreement on European productivity levels.

MODEL

We study a panel of Western European nations in the period 1985 – 2007. As the Schengen agreement was gradually introduced to the countries in Western Europe, we exploit the staggered rollout of Schengen membership to assess the effect the agreement has had on Eastern European levels of labor productivity. In 1995, the agreement covered France, Germany, Spain, Portugal, The Netherlands, Belgium and Luxembourg. In 1997, It was expanded to Italy and Austria. In 2000, Greece was included, and in 2001, the Nordic countries (Sweden, Norway, Denmark, Finland and Iceland) joined the agreement (European Commission, 2025b)



Figure 2: Timeline of countries joining the Schengen Agreement

We utilize an Event study design with a staggered rollout process, as outlined by Callaway & Sant’Anna (2021). We implement this design by exploiting that the agreement did not enter into force for all participating countries simultaneously. The control group is therefore composed of the not yet treated countries (NYTs). This approach can in some cases introduce bias through anticipatory effects. We do, however, argue that these effects are highly limited in this very case due to the nature of the “treatment”. As border controls are not fully abolished until a given country officially joins the agreement, most anticipatory effects are naturally controlled for. One could argue that because all participating states signed the agreement itself on average five years before entry, people and firms prepare for the lifting of border checks by making pre-treatment adaptations to their practices, which itself could be argued to cause increases in productivity.

We show in the appendix that using the signing date and not the entry date as the treatment point, increases in productivity (measured by GDP per worker) only emerge around the time of implementation, approximately five years after signing. This pattern indicates that firms and individuals did not significantly adjust their behavior prior to enforcement, and that most productivity gains occur once border controls are lifted. Consequently, anticipatory effects are likely negligible in this case.

Furthermore, we have also chosen to exclude all former eastern-bloc countries from the analysis to rule out the possibility of catch-up bias due to productivity gains originating from the abolition of communism. We also stop the analysis in the year 2007, as this prevents potential bias created by the 2008 global financial crisis to act as a confounder. Validity concerns will be further assessed in the discussion section.

We use the Penn World Tables dataset (Feenstra et al. 2015) to assess the potential productivity gains obtained through the implementation of the Schengen agreement. As we cluster on the country level, the given timeline we study only provides us with 15 clusters. This can be very problematic as shown by Baker et al. (2025). To mitigate bias caused by a low number of clusters, we use a multiplier bootstrap. This increases finite sample coverage by resampling at the country level and preserving within-country dependence. For our main outcome, $Y_{it}^{(w)} = \log \text{GDP per worker}$, this lowers the estimated p -value from 2.5% to 2.1%. The non-bootstrapped estimates are shown in the appendix.

We study effects on labor productivity through two outcome variables: $Y_{it}^{(h)} = \log \text{GDP per hour}$ and $Y_{it}^{(w)} = \log \text{GDP per worker}$ (Both PPP adjusted using 2017 prices). We predict that GDP per worker

will have increased as discussed earlier. As a secondary outcome, we study how the implementation of the agreement has affected investments as a share of real GDP (PPP adjusted using 2005 prices): $Y_{it}^{(ki)} = \text{Investments}/\text{GDP}$.

In our model, we let i denote countries and t denote years. We let G_i be the year that country i became a part of the agreement. Treatment status is therefore defined as $D_{it} = 1\{t \geq G_i\}$. Furthermore, as the year of entry differs across countries, we re-index time so that “0” is the country’s own entry year $R_{it} \equiv t - G_i$. R_{it} will therefore take a negative value for any number of years before the entry of country i . For any given year t , the control group consists of the not-yet-treated countries: $c^{NYT} = \{j: t < G_j\}$. With this design, we avoid the potential biases introduced by using the two-way fixed effects estimator, as we do not compare treated countries to already treated countries.

If we let $Y_{it}(g)$ denote the outcome we would observe at time t if country i entered the agreement in year g and $Y_{it}(\infty)$ be the outcome if country i never entered the agreement (in our timeframe), the group-time ATT for the cohort that actually enters in year g is:

$$ATT(g, t) = E[Y_{it}(g) - Y_{it}(\infty) | G_i = g], t \geq g$$

We therefore establish a clear counterfactual between countries that are a part of Schengen and countries not yet treated in year t . We express event time $e = t - g$ and average across all cohorts that have support at e . Support in this case means that the calendar year $t = g + e$ is included in the sample and that there exists a not yet treated control group at that given t . The weight coefficient $w_{g,e}$ reflects the share of observations for a given cohort treated e years after it entered the agreement. This is equivalent to the share of treated units belonging to a given cohort for a given e . As all four cohorts enter into the treated state (Schengen membership), $w_{g,e}$ accumulates to 1.

$$ATT(e) = \sum_g w_{g,e} ATT(g, g + e)$$

Our model builds upon several key assumptions that need to be validated for any estimates produced by the model to be interpreted causally. Firstly, there must be no anticipation effects. As discussed earlier in this section, we do not believe that anticipation effects can play a major role in this case due to the very nature of the treatment (any potential gains cannot be fully realized before border controls are actually abolished). Furthermore, any causal inference necessitates that the parallel trends assumption holds. In our case, this translates to parallel trends in untreated outcomes. Without the agreement, the countries that joined should have evolved similarly to those who did not. Although this assumption is not directly testable, we assess whether the pre-treatment trends are parallel in the following section.

RESULTS

For $Y_{it}^{(h)} = \log GDP \text{ per hour}$, the introduction of the Schengen agreement has had no statistically significant effects. The pre-trend assumption appears to hold as all confidence intervals for the pre-treatment group include 0 for all periods. The effects in the post-treatment are, however, non-significant although the model estimates some increases in $\log GDP \text{ per hour}$. This will be further discussed in the following section.

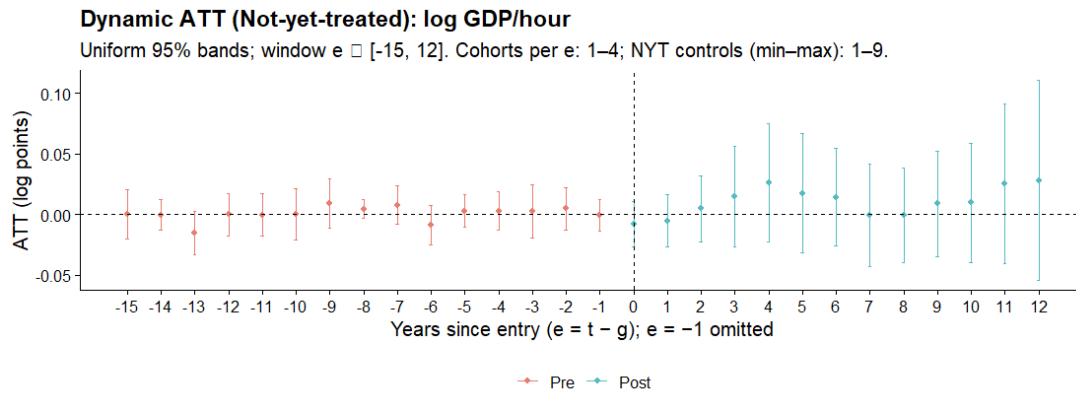


Figure 3: Effect of the Schengen Agreement on Log GDP per hour (PPP adjusted)

For $Y_{it}^{(w)} = \log GDP \text{ per worker}$, effects appear to be significant post-treatment. The parallel pre-trend assumption also holds in this case, as the confidence intervals for all pre-treatment groups include 0 for all periods. The treatment effect in this case, however, is statistically significant in contrast to the other main outcome we study. This result also aligns with our earlier prediction of increased labor productivity. The fact that the effect is only significant after a minimum of 5 years after treatment shows that people and firms do not immediately adapt to the reform but instead need a considerable amount of time to take the changes into account. This pattern also suggests that there are few if any anticipatory effects, as adaptation to the reform happens after it is implemented and not before.

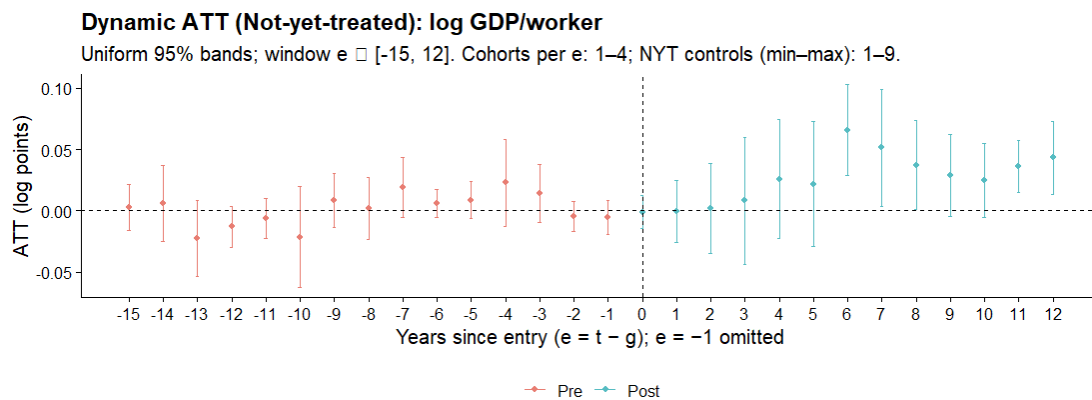


Figure 4: Effect of the Schengen agreement on Log GDP per worker (PPP adjusted)

Numerically, we estimate that the Schengen agreement led to a 1% (0.01 log point) increase in GDP per hour. This estimate, however, had an estimated p-value of 38,2%, indicating that it is nowhere near

statistically significant. This aligns with the graphical interpretation of the effect from figure 1. For GDP per worker, we estimate that the agreement led to a 2.4% (0.024 log point) increase. The p-value of 2.1% indicates that the estimate is statistically significant at the 5% level. To obtain this estimate, we use a country-clustered multiplier bootstrap which provides us with better calibrated *SEs* with few clusters. As shown in the table below and in the appendix, the bootstrapped *SE* for the overall ATT is slightly smaller than the asymptotic clustered *SE*. The *p*-value for $Y_{it}^{(w)} = \log \text{GDP per worker}$ moves from 2.5% to 2.1%. The point estimate is, however, unchanged.

Overall ATT, Callaway & Sant'Anna (NYT controls)					
outcome	ATT (log)	SE	95% CI	ATT (%)	p-value
log GDP/hour	0.010	(0.011)	[-0.012, 0.032]	1.0	0.382
log GDP/worker	0.024*	(0.010)	[0.004, 0.044]	2.4	0.021

Figure 5: Estimated ATT of the Schengen agreement on our two selected main outcome variables

For our secondary outcome, investment share of GDP, we estimate an ATT of 1.85 percentage points while employing the same multiplier bootstrapping method as for our main outcomes. This estimate is strongly significant with $p < 0.001$. The parallel pre-trends assumption also appears to hold in this case. This result suggests that the implementation of the Schengen agreement led to an increase in investments as share of GDP. The result should be interpreted with some caution as the Dot-com bubble of the early 2000's may have deflated GDP, resulting in an overestimation of the effect in our model. However, as the effect magnifies over time in our dynamic ATT plot, the overall causal effect still appears to be positive. In the following section, we provide an interpretation of this result and discuss it in the light of the theoretical frameworks we presented earlier.

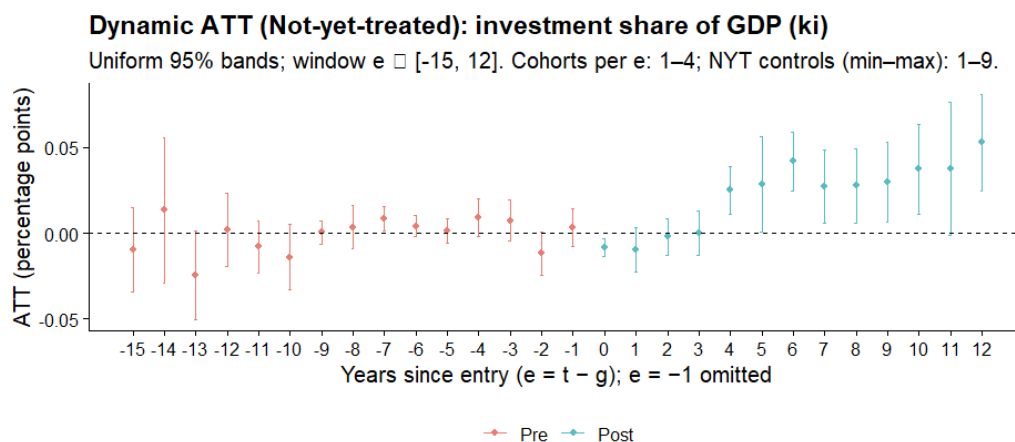


Figure 6: Effect of the Schengen agreement on investment share of GDP (PPP adjusted)

Overall ATT, Callaway & Sant'Anna (NYT controls)				
Outcome	ATT (pp)	SE (pp)	95% CI	p-value
Investment share of GDP (ki)	1.854***	(0.441)	[0.990, 2.718]	<0.001

Figure 7: Estimated ATT of the Schengen agreement on investment/GDP (PPP-adjusted)

DISCUSSION AND CONCLUSION

Our results partially align with the predictions we made earlier. As discussed in the literature review, increased openness between nations is predicted to lead to increases in overall levels of welfare for all countries taking part in the exchange of goods and services. Furthermore, our secondary analysis suggests that the increase in trans-European mobility have increased investments as a share of real GDP for our sample of countries, with existing literature suggesting two key mechanisms that explain this pattern: The increased labor mobility incentivizes firms to scale up operations where they are the most productive, necessitating investments in the necessary infrastructure needed to perform the upscaling (Melitz, 2003; Bernard et al. 2006). Additionally, the lower costs of coordination due to the abolishment of border controls raise the payoff to capital deployment, of which a share of the increased returns are reinvested in further capital accumulation, thereby driving an increase in Investment/GDP (Helpman et al, 2004; Bernard et al, 2011). Overall, our results therefore appear to be in line with economic theory.

In the aftermath of the 2015 European refugee crisis, several countries have suspended the Schengen agreement to better monitor the flow of people across their borders (European Commission, 2025). Although the re-institution of border controls is intended to be temporary, they still set a damaging precedent by justifying the suspension of a policy tool that has created benefits for the whole continent over the last 30 years. According to our findings, this is also likely to have adverse implications for European productivity levels in a time where Europe struggles with innovation and competitiveness relative to its main geopolitical rival: China.

Although we provide a significant and positive estimate on one of our select measures for labor productivity, we have made tradeoffs that introduce several potential weaknesses into our research design. Firstly, the choice to exclude the whole of Eastern Europe severely limits the external validity of our findings. As we only estimate the effects of the agreement on Western and Northern European countries, the coefficient estimates do likely not represent the effect the agreement has had on labor productivity in other European countries. Another potential weakness of our design is that we do not account for geographic heterogeneity. This could potentially be important as peripheral countries such as Finland or Portugal likely have benefited less than more centrally located countries like France, Germany and the Netherlands.

Secondly, our choice to stop the analysis in 2007 in practice only lets us study the effect of the agreement for its first 12 years of existence. One could argue that external events that took place in the period 2007 - 2025 have altered the effects that the agreement has had on labor productivity. We still chose to limit our analysis to the first 12 years of the agreement for two main reasons: As most of Eastern Europe entered Schengen in December 2007, we fear that including it would bias our estimate upward because of the continued catch-up from communism. Secondly, we argue that the Global Financial Crisis would have acted as a major confounder as it has had long-term adverse effect on much of Europe. We therefore

argue that although our design has several flaws, it still provides an accurate assessment of the impact of the Schengen agreement on Western European labor productivity in the period 1995 – 2007.

Thirdly, our analysis may be affected by potential spillover effects across countries. Productivity gains in countries that joined Schengen could influence neighboring countries that had not yet joined, for example through cross-border commuting or trade integration. Such spillovers violate the assumption that untreated countries serve as a valid counterfactual and may lead to an under or overestimation of the true effect of Schengen on labor productivity. While our staggered rollout design mitigates some of this concern by comparing each treated country to not-yet-treated peers, we cannot completely rule out that these indirect effects influence our estimates. Therefore, our results should be interpreted with caution as potentially capturing only a portion of the overall productivity impact of the agreement.

We conclude that the Schengen agreement has clearly benefited participating nations. The ability to freely cross borders without border controls has led to further European integration, and the cementing the EU's 4 freedoms in the process. The analysis we provide illustrates a positive effect on European labor productivity, suggesting an increase in GDP per worker increased by 2.4 percent as a direct result. The trend towards increasing mistrust among European countries is thus likely to affect productivity and competitiveness negatively at a time when most countries on the continent are already struggling.

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APPENDIX

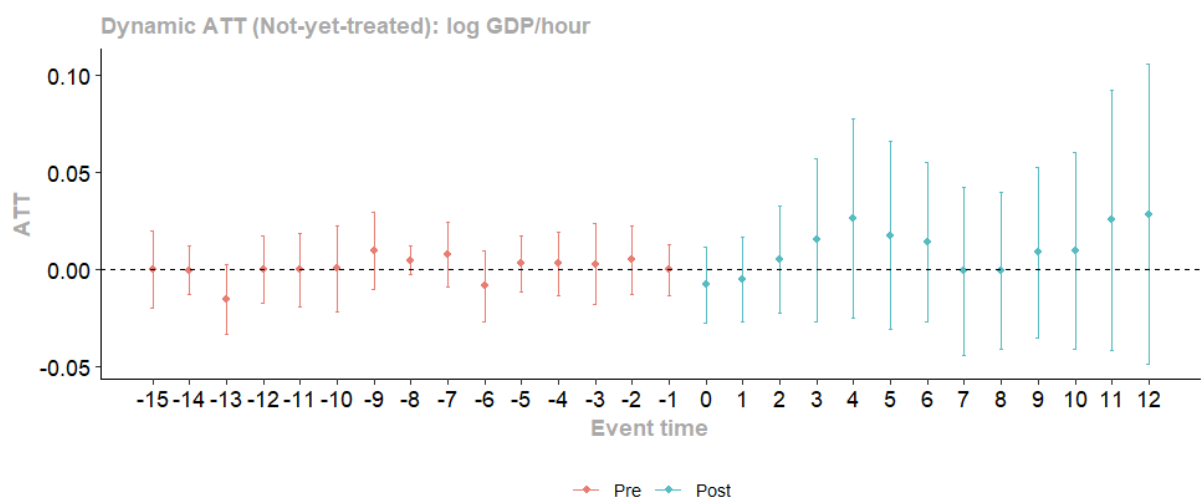


Figure 8: Effect of the Schengen Agreement on Log GDP per hour (PPP adjusted) (non-bootstrapped)

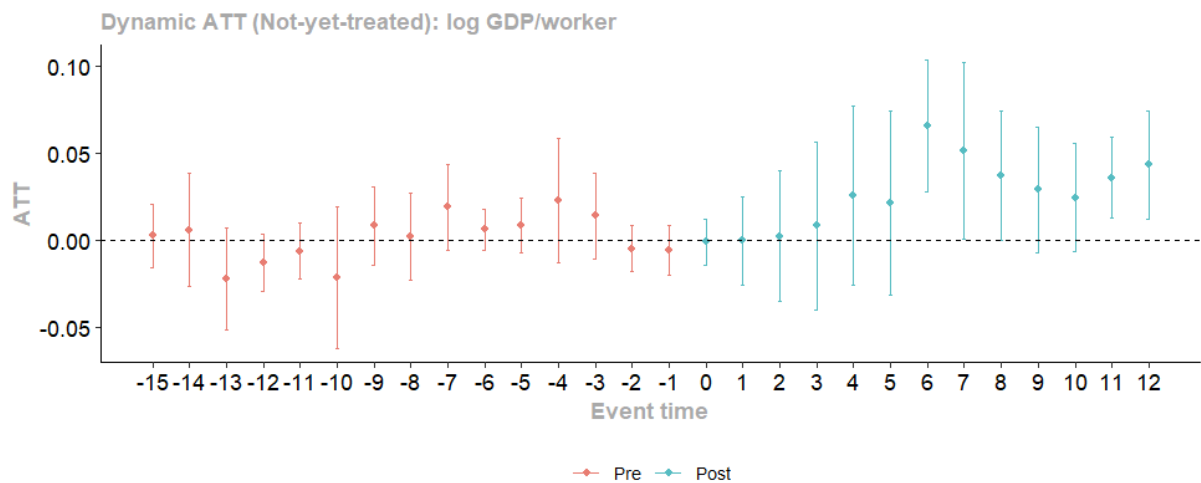


Figure 9: Effect of the Schengen Agreement on Log GDP per worker (PPP adjusted) (non-bootstrapped)

Overall ATT, Callaway & Sant Anna (NYT controls)					
Outcome	ATT (log)	SE	95% CI	ATT (%)	p-value
log GDP/hour	0.010	(0.011)	[-0.013, 0.032]	1.0	0.394
log GDP/worker	0.024* *	(0.011)	[0.003, 0.045]	2.4	0.025

Figure 10: Estimated overall effects of the Schengen agreement on our two selected main outcome variables (non-bootstrapped)

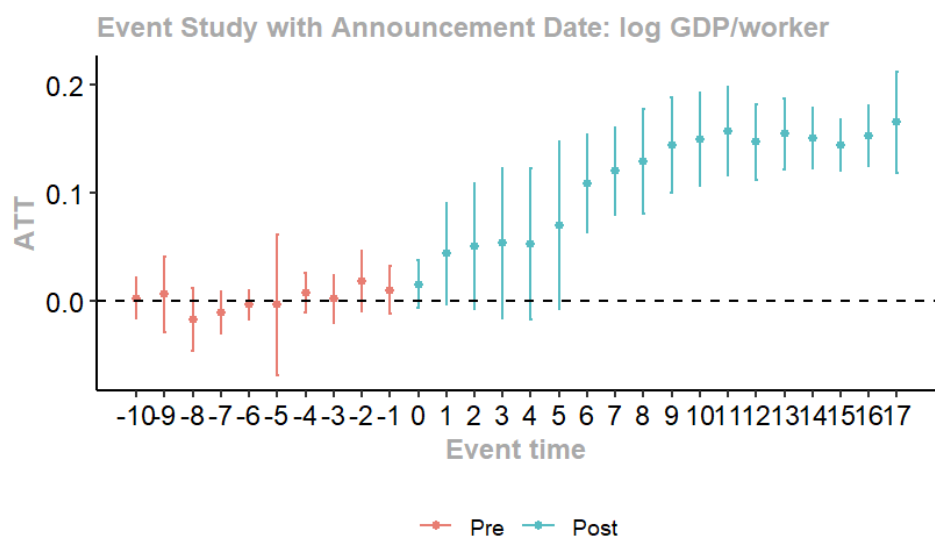


Figure 11: Robustness check: Using signing date as a proxy for entry to control for anticipatory effects on GDP per worker (PPP)